Classroom Activity 1: Choose a Laptop for the Lab Using AW

Annual Worth (AW) Analysis

Objective. Your computer engineering lab needs new laptops. Three models are available. Use Annual Worth (AW) analysis to determine which laptop is the most economical option.

Assume: MARR = 10%.

Laptop Options

Laptop A — Basic Dev Machine

- Purchase price: $1,200
- Annual maintenance: $150
- Battery replacement: $40 per year
- Life: 3 years
- Salvage value: $150

Laptop B — Performance Model

- Purchase price: $1,600
- Annual maintenance: $120
- Battery replacement: $80 every 2 years
- Life: 4 years
- Salvage value: $250

Laptop C — High-End Engineering Laptop

- Purchase price: $2,200
- Annual maintenance: $90
- Battery replacement: $50 per year

- Life: 5 years
- Salvage value: \$300

Instructions

Step 1. Compute capital recovery (CR):

$$CR = -P(A/P, 10\%, n) + S(A/F, 10\%, n)$$

Step 2. Add annual operating costs:

$$AW = CR + (\text{maintenance}) + (\text{battery cost})$$

Step 3. Compare AW values and choose the laptop with the lowest AW.

Student Work Area

Laptop A

$$CR_A = _____$$
$$AW_A = _____$$

Laptop B

$$CR_B = _____$$
$$AW_B = _____$$

Laptop C

$$CR_C = _____$$
$$AW_C = _____$$

Answer Key

Annual Worth (AW) Calculations

General Factor Formulas (at interest rate i and n years):

$$(A/P, i, n) = \frac{i(1+i)^n}{(1+i)^n - 1}, \qquad (A/F, i, n) = \frac{i}{(1+i)^n - 1}, \qquad (F/P, i, n) = (1+i)^n$$

Useful Numerical Factors at 10%:

$$(A/P, 10\%, 3) = 0.4021, \quad (A/F, 10\%, 3) = 0.3014$$
$$(A/P, 10\%, 4) = 0.3155, \quad (A/F, 10\%, 4) = 0.2155$$
$$(A/P, 10\%, 5) = 0.2638, \quad (A/F, 10\%, 5) = 0.1638$$

Laptop A

Capital recovery:

$$CR_A = -P(A/P, 10\%, 3) + S(A/F, 10\%, 3) = -1{,}200(A/P, 10\%, 3) + 150(A/F, 10\%, 3)$$
$$CR_A = -1{,}200(0.4021) + 150(0.3014) = -482.52 + 45.21 = -437.31$$

Annual worth:

$$AW_A = CR_A - \text{maintenance} - \text{battery cost} = -437.31 - 150 - 40 = \boxed{-627.31}$$

Laptop B

Battery cost averaged over 2 years:

$$\text{Battery per year} = \frac{80}{2} = 40$$

Capital recovery:

$$CR_B = -P(A/P, 10\%, 4) + S(A/F, 10\%, 4) = -1{,}600(A/P, 10\%, 4) + 250(A/F, 10\%, 4)$$
$$CR_B = -1{,}600(0.3155) + 250(0.2155) = -504.80 + 53.88 = -450.92$$

Annual worth:

$$AW_B = CR_B - \text{maintenance} - \text{battery cost} = -450.92 - 120 - 40 = \boxed{-610.92}$$

Laptop C

Capital recovery:

$$CR_C = -P(A/P, 10\%, 5) + S(A/F, 10\%, 5) = -2{,}200(A/P, 10\%, 5) + 300(A/F, 10\%, 5)$$

$$CR_C = -2{,}200(0.2638) + 300(0.1638) = -580.36 + 49.14 = -531.22$$

Annual worth:

$$AW_C = CR_C - \text{maintenance} - \text{battery cost} = -531.22 - 90 - 50 = \boxed{-671.22}$$

Conclusion

The lowest annual worth is:

$$AW_B = -\$610.92$$

Laptop B is the most economical choice.